Discreteness of sets generated by fixed-parameter extrapolation

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Abstract

Starting from 0 and 1, repeatedly apply the operation $(a, b) \mapsto (1-\lambda)a + \lambda b$, with a fixed complex parameter λ. We prove that the resulting set is discrete exactly when λ is an algebraic integer whose other conjugates, apart from λ and $\overline{\lambda}$, are real and lie in $(0, 1)$. This gives the converse to the known sufficient condition for separated extrapolation sets. The proof adjusts integer polynomials within a residue class while preserving their values at λ. If there is an additional conjugate outside $(0, 1)$, density of the corresponding field projections then produces nonzero generated points arbitrarily close to 0.

1 The discreteness question

For $\lambda \in \mathbb{C}$, write

$$a \star_\lambda b = (1-\lambda)a + \lambda b,$$

and let Q_λ be the smallest subset of $\mathbb{C}$ containing $0, 1$ and closed under $\star_\lambda$. Each point of Q_λ is obtained by finitely many operations; no topological closure is taken. The operation has been used to generate point sets in aperiodic order, where separation distinguishes discrete patterns from arbitrarily fine ones [1].

Following Fenner, Green and Homer [1, Definition 4.1], an algebraic integer λ is *strong PV* if every conjugate other than λ and $\overline{\lambda}$ is real and belongs to $(0, 1)$. Integers and nonreal quadratic algebraic integers satisfy this condition vacuously. Strong PV parameters give uniformly discrete sets [1, Theorem 4.12]; discreteness also forces algebraic integrality [2, Theorem 8.2]. Campbell recently proved the necessary conjugate restriction for totally real algebraic integers [3, Theorem 2]. The full complex-parameter converse is Conjecture 11.1 of [1].

Theorem 1. *For every $\lambda \in \mathbb{C}$, the set Q_λ is discrete if and only if λ is strong PV. In that case,*

$$|x - y| \geq 1 \qquad (x, y \in Q_\lambda,\ x \neq y).$$

The main issue is the converse. A generated point can be represented by an integer polynomial that stays between 0 and 1 on the real unit interval. We show that the interval condition can be imposed after fixing the polynomial's residue modulo the minimal polynomial of λ. An extra conjugate outside $(0, 1)$ leaves an unbounded field direction; projecting away that direction gives enough small residues to contradict discreteness. The argument treats real parameters with nonreal conjugates and nonreal parameters of arbitrary degree together.

2 Integer polynomials and prescribed residues

Let $Q_{[t]} \subseteq \mathbb{Z}[t]$ be generated by $0, 1$ under $(f, g) \mapsto (1-t)f + tg$. Fenner, Green and Homer proved [1, Theorem 3.6 and Corollary 3.8] that

$$Q_{[t]} = \{0, 1\} \cup \{P \in \mathbb{Z}[t] : 0 < P(u) < 1 \text{ for every } 0 < u < 1\}, \tag{1}$$

$$Q_\lambda = \{P(\lambda) : P \in Q_{[t]}\}. \tag{2}$$

These are statements about finite generation. In particular, a polynomial in the right-hand side of (1) gives a finite operation tree for its value at every complex parameter.

Put $q(t) = t(1-t)$. We will need the elementary integer approximation property behind rounded Bernstein polynomials; see [4, Section 1] for this classical construction.

Lemma 2. *If $h \in C([0,1], \mathbb{R})$ and $h(0), h(1) \in \mathbb{Z}$, then h can be uniformly approximated by polynomials $H \in \mathbb{Z}[t]$ with $H(0) = h(0)$ and $H(1) = h(1)$.*

Proof. For $n \geq 1$, round each coefficient $\binom{n}{k} h(k/n)$, $1 \leq k < n$, to an integer c_k with error at most $1/2$. Set $c_0 = h(0)$, $c_n = h(1)$ and

$$H_n(t) = \sum_{k=0}^{n} c_k t^k (1-t)^{n-k} \in \mathbb{Z}[t].$$

The endpoints are exact. If $B_n h$ is the usual Bernstein polynomial,

$$|H_n(t) - B_n h(t)| \leq \frac{1}{2} \sum_{k=1}^{n-1} t^k (1-t)^{n-k}.$$

For $0 \leq t \leq 1/4$, the sum is at most

$$\frac{t(1-t)^n}{1-2t} \leq 2t(1-t)^n \leq \frac{2}{n+1}.$$

The same bound holds on $[3/4, 1]$ by symmetry. On $[1/4, 3/4]$ the sum is at most $(n-1)(3/4)^n$. Thus the rounding error tends uniformly to zero.

For completeness, the weights $w_k = \binom{n}{k} t^k (1-t)^{n-k}$ satisfy

$$\sum_k w_k = 1, \qquad \sum_k (k/n - t)^2 w_k = \frac{t(1-t)}{n} \leq \frac{1}{4n},$$

by the binomial identities for the first two moments. If $\omega_h(\rho)$ denotes the modulus of continuity of h, splitting the sum at $|k/n - t| = \rho$ gives

$$|B_n h(t) - h(t)| \leq \omega_h(\rho) + \frac{\|h\|_\infty}{2n\rho^2}.$$

First choose ρ small and then n large. Hence $B_n h \to h$ uniformly, and the same is true of H_n. □

Lemma 3. *Let $m, f \in \mathbb{Z}[t]$ with $m(0)m(1) \neq 0$. Suppose that*

$$0 < q(\mu) f(\mu) < 1$$

at every real zero μ of m in $(0,1)$. There is an $H \in \mathbb{Z}[t]$ such that $P = q(f + mH)$ satisfies $0 < P(t) < 1$ for all $0 < t < 1$. For every zero λ of m,

$$P(\lambda) = q(\lambda) f(\lambda) \in Q_\lambda.$$

Proof. We first choose a continuous real function h such that

$$F(t) := f(t) + m(t)h(t) > 0, \qquad q(t)F(t) < 1 \quad (0 \le t \le 1), \tag{3}$$

with integer values of h at the endpoints. At an interior real zero of m, the constant $h = 0$ satisfies (3) on a neighborhood. At an endpoint $e \in \{0, 1\}$, choose $n_e \in \mathbb{Z}$ with $f(e) + m(e)n_e > 0$. Since $q(e) = 0$, the constant $h = n_e$ works on a neighborhood of e. At any remaining $t_0 \in (0, 1)$, the constant

$$c_{t_0} = \frac{1/(2q(t_0)) - f(t_0)}{m(t_0)}$$

makes $F(t_0) = 1/(2q(t_0))$, so it also works locally.

Choose the interior neighborhoods to avoid the endpoints and each endpoint neighborhood to avoid the opposite endpoint. A finite subcover of $[0, 1]$ has a continuous nonnegative partition of unity. For example, normalize the distances to the complements of its relative-open members. Take h to be the corresponding convex combination of the local constants. The inequalities in (3) are strict affine inequalities in $h(t)$, so they are preserved. At each endpoint only its designated neighborhood contributes, giving $h(e) = n_e$.

Both continuous functions F and $1 - qF$ have positive minima, say a and b. With $M = \max(1, \|m\|_\infty)$, Lemma 2 gives $H \in \mathbb{Z}[t]$ satisfying

$$\|H - h\|_\infty < \frac{\min(a, b)}{2M}.$$

Then $|m(H - h)| < \min(a, b)/2$. Since $0 \le q \le 1/4$ on $[0, 1]$, replacing h by H preserves both inequalities. Thus $0 < q(f + mH) < 1$ on $(0, 1)$. Finally, $m(\lambda) = 0$ gives the stated value, and (1)–(2) put it in the finite generated set Q_λ. □

3 Density after omitting a conjugate

We use a character test for additive subgroups of a real Euclidean space. Its short proof is included to make the density argument explicit.

Lemma 4. *If an additive subgroup $G \subseteq \mathbb{R}^N$ is not dense, there is a nonzero real linear functional L with $L(G) \subseteq \mathbb{Z}$.*

Proof. Let $A = \overline{G}$ and choose a vector subspace $V \subseteq A$ of maximal dimension. Orthogonal projection gives $A = V + D$, where $D = A \cap V^\perp$. The subgroup D is discrete. Otherwise some nonzero $d_j \in D$ tend to 0, and a subsequence of $d_j/|d_j|$ tends to a unit vector $v \in V^\perp$. For every $u \in \mathbb{R}$ choose integers k_j with $k_j|d_j| \to u$. Closedness gives $k_j d_j \to uv \in D$, so $V + \mathbb{R}v \subseteq A$, contrary to maximality.

Put $W = \operatorname{span}_{\mathbb{R}} D$. If $W \neq V^\perp$, take a nonzero functional annihilating $V + W$. Otherwise choose a real basis $d_1, \dots, d_r$ of $V^\perp$ from D and let $D_0 = \sum_i \mathbb{Z}d_i$. A bounded fundamental parallelepiped for D_0 meets the discrete group D in finitely many points, so D/D_0 is finite. Some positive integer J satisfies $JD \subseteq D_0$. The functional with values $J, 0, \dots, 0$ on $d_1, \dots, d_r$, extended by zero on V, takes integer values on A. It is nonzero: $r > 0$ because A is proper. Either case gives the required L. □

Let λ be an algebraic integer of degree d, $K = \mathbb{Q}(\lambda)$ and $O = \mathbb{Z}[\lambda]$. List the real embeddings as $\sigma_1, \dots, \sigma_r$, and choose one representative τ_j from each of the s nonreal conjugate pairs. The full embedding space and map are

$$E = \mathbb{R}^r \times \mathbb{C}^s, \qquad r + 2s = d, \qquad \Sigma(\alpha) = (\sigma_i(\alpha), \tau_j(\alpha)).$$

A coordinate selection T keeps some real factors and some *whole* complex factors. Write $\pi_T : E \to E_T$ for the projection.

Proposition 5. *If T is a proper coordinate selection, then $\pi_T\Sigma(O)$ is dense in E_T, viewed as a real Euclidean space.*

Proof. The empty selection is trivial. Otherwise, if the image is not dense, Lemma 4 gives a nonzero real functional L on E_T taking integer values on that image. Let $\widetilde{L} = L \circ \pi_T$. Since $1, \lambda, \dots, \lambda^{d-1}$ is a rational basis of K, nondegeneracy of the trace pairing gives a unique $b \in K$ such that

$$\mathrm{Tr}_{K/\mathbb{Q}}(b\lambda^j) = \widetilde{L}(\Sigma(\lambda^j)) \quad (0 \le j < d).$$

Indeed, the right-hand sides are integers, and the pairing is nondegenerate because $b \neq 0$ implies $\mathrm{Tr}(bb^{-1}) = d \neq 0$.

On E this trace is represented by the real functional

$$T_b((x_i), (z_j)) = \sum_i \sigma_i(b)x_i + 2\,\mathrm{Re}\sum_j \tau_j(b)z_j. \tag{4}$$

The vectors $\Sigma(1), \dots, \Sigma(\lambda^{d-1})$ form a real basis of E. A real linear relation would give a real polynomial of degree $< d$ vanishing at all real roots and at one root in every nonreal pair, hence at their conjugates as well. The d roots of the minimal polynomial are distinct, so the relation must be zero.

Thus $\widetilde{L} = T_b$ on E. But $\widetilde{L}$ vanishes on an omitted coordinate. By (4), the corresponding real or complex embedding of b is zero; for a complex factor, $z \mapsto 2\,\mathrm{Re}(cz)$ vanishes identically only if $c = 0$. Field embeddings are injective, so $b = 0$ and then $L = 0$, a contradiction. □

4 Proof of the characterization

Proof of Theorem 1. Suppose first that Q_λ is discrete. The integrality theorem [2, Theorem 8.2] makes λ an algebraic integer. If $\lambda = 0$ or 1, it is already strong PV. If $0 < \lambda < 1$, the nonzero points $\lambda^n \in Q_\lambda$ tend to 0, a contradiction. We may therefore assume $\lambda \notin [0, 1]$.

Let $m \in \mathbb{Z}[t]$ be its monic minimal polynomial, and list its real roots in $(0, 1)$ as $\mu_1, \dots, \mu_k$. Irreducibility gives $m(0)m(1) \neq 0$. In the full embedding space retain the physical coordinate corresponding to λ, together with all these k real coordinates. Choose the representative of the physical complex pair to be λ itself when λ is nonreal. The resulting space is

$$E_T = F \times \mathbb{R}^k, \qquad F = \begin{cases} \mathbb{R}, & \lambda \in \mathbb{R}, \\ \mathbb{C}, & \lambda \notin \mathbb{R}. \end{cases}$$

If λ is not strong PV, a conjugate other than $\lambda, \overline{\lambda}$ lies outside the real interval $(0, 1)$. It belongs to an omitted real factor or an omitted complex pair. Hence T is proper, and Proposition 5 applies. For any $\varepsilon > 0$, density gives $\alpha = f(\lambda) \in O$, $f \in \mathbb{Z}[t]$, with its selected coordinates in the nonempty open set

$$\left\{ z \in F : \frac{\varepsilon}{2|q(\lambda)|} < |z| < \frac{\varepsilon}{|q(\lambda)|} \right\} \times \prod_{i=1}^{k} \left(\frac{1}{4q(\mu_i)}, \frac{3}{4q(\mu_i)} \right). \tag{5}$$

Here $q(\lambda) \neq 0$; the empty product is allowed. Thus $\alpha \neq 0$, $|q(\lambda)\alpha| < \varepsilon$, and $0 < q(\mu_i)f(\mu_i) < 1$ for every i. Lemma 3 puts the nonzero number $q(\lambda)\alpha$ in Q_λ. Since ε was arbitrary, 0 is an accumulation point, contradicting discreteness. This proves the necessary conjugate restriction.

Conversely let λ be strong PV and take distinct $x, y \in Q_\lambda$. Tree induction gives $a = x - y \in \mathbb{Z}[\lambda]$. At every other real conjugate $\mu \in (0,1)$ the evaluated trees stay in $[0,1]$, so $|\sigma_\mu(a)| \le 1$. The norm of the nonzero algebraic integer a is a nonzero integer. Its physical factor is a in the real case and $a\overline{a}$ in the nonreal case. Therefore

$$1 \le |\,\mathrm{N}_{K/\mathbb{Q}}(a)| \le \begin{cases} |a|, & \lambda \in \mathbb{R}, \\ |a|^2, & \lambda \notin \mathbb{R}. \end{cases}$$

This gives $|x - y| \ge 1$. For $\lambda = 0, 1$ the set is exactly $\{0, 1\}$, so the same separation holds. □

AI use. The proof and exposition were developed with assistance from OpenAI Codex, including source retrieval and checks of the argument. Those preparation checks do not constitute independent human review. Independent mathematical review of the submitted argument is requested.

References

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